

identification and parameter estimation in fluids that are precious or otherwise difficult to study in bulk. [1] K. D. Housiadas and R. I. Tanner, A high-order perturbation solution for the steady sedimentation of a sphere in a viscoelastic fluid, *J. Non-Newton. Fluid Mech.* 233, 166 (2016). [2] G. McKinley, Steady transient motion of spherical particles in viscoelastic liquids, in *Transp. Process. Bubbles, Drops, Part.*, edited by E. Chhabra and D. D. Kee (Taylor & Francis, 2001) 2nd ed. [3] K. D. Housiadas and R. I. Tanner, Perturbation solution for the viscoelastic 3D flow around a rigid sphere subject to simple shear, *Phys. Fluids* 23, 10.1063/1.3615518 (2011). [4] H. Giesekus, Die simultane Translations- und Rotationsbewegung einer Kugel in einer elastoviskosen Flüssigkeit, *Rheol. Acta* 3, 59 (1963). [5] K. D. Housiadas and R. I. Tanner, The angular velocity of a freely rotating sphere in a weakly viscoelastic matrix fluid (2011). [6] K. D. Housiadas and R. I. Tanner, Perturbation solution for the viscoelastic flow around a rigid sphere under pure uniaxial elongation, *J. Non-Newton. Fluid Mech.* 167-168, 75 (2012). [7] S. D. Gkormpatsis, E. A. Gryparis, K. D. Housiadas, and A. N. Beris, Steady sphere translation in a viscoelastic fluid with slip on the surface of the sphere, *J. Non-Newton. Fluid Mech.* 275, 104217 (2020). [8] L. E. Becker, G. H. McKinley, and H. A. Stone, Sedimentation of a sphere near a plane wall: Weak non-Newtonian and inertial effects, *J. Non-Newton. Fluid Mech.* 63, 201 (1996). [9] C. Bodart and M. J. Crochet, The time-dependent flow of a viscoelastic fluid around a sphere, *J. Non-Newton. Fluid Mech.* 10.1016/0377-0257(94)80029-4 (1994). [10] L. E. Becker, G. H. McKinley, H. K. Rasmussen, and O. Hassager, The unsteady motion of a sphere in a viscoelastic fluid, *J. Rheol.* 38, 377 (1994). [11] M. Chilcott and J. Rallison, Creeping flow of dilute polymer solutions past cylinders and spheres, *J. Non-Newton. Fluid Mech.* 29, 381 (1988). [12] M. N. Moore and M. J. Shelley, A weak-coupling expansion for viscoelastic fluids applied to dynamic settling of a body, *J. Non-Newton. Fluid Mech.* 183-184, 25 (2012). [13] A. M. Mollinger, E. C. Cornelissen, and B. H. Van Den Brule, An unexpected phenomenon observed in particle settling: Oscillating falling spheres, *J. Non-Newton. Fluid Mech.* 10.1016/S0377-0257(98)00187-6 (1999). [14] Y. Zhang and S. J. Muller, Unsteady sedimentation of a sphere in wormlike micellar fluids, *Phys. Rev. Fluids* 3, 10.1103/PhysRevFluids.3.043301 (2018). [15] L. G. Wilson and W. C. Poon, Small-world rheology: an introduction to probe-based active microrheology, *Phys. Chem. Chem. Phys.* 13, 10617 (2011). [16] P. Dhar, Y. Cao, T. M. Fischer, and J. A. Zasadzinski, Active interfacial shear microrheology of aging protein films, *Phys. Rev. Lett.* 104, <https://doi.org/10.1103/PhysRevLett.104.016001> (2010). [17] M. L. Gardel, M. T. Valentine, and D. A. Weitz, *Microrheology*, *Microscale Diagnostic Tech.*, 1 (2005). [18] M. Guo, A. J. Ehrlicher, S. Mahammad, H. Fabich, M. H. Jensen, J. R. Moore, J. J. Fredberg, R. D. Goldman, and D. A. Weitz, The role of vimentin intermediate filaments in cortical and cytoplasmic mechanics, *Biophys. J.* 10.1016/j.bpj.2013.08.037 (2013). [19] T. M. Squires and T. G. Mason, Fluid Mechanics of Microrheology, *Annu. Rev. Fluid Mech.* 10.1146/annurev-fluid-121108-145608 (2010). [20] R. M. Robertson-Anderson, Optical Tweezers Microrheology: From the Basics to Advanced Techniques and Applications, *ACS Macro Lett.* 7, 968 (2018). [21] A. Meyer, A. Marshall, B. G. Bush, and E. M. Furst, Laser tweezer microrheology of a colloidal suspension, *J. Rheol.* 50, 77 (2006). [22] R. R. Brau, J. M. Ferrer, H. Lee, C. E. Castro, B. K. Tam, P. B. Tarsa, P. Matsudaira, M. C. Boyce, R. D. Kamm, and M. J. Lang, Passive and active microrheology with optical tweezers, *J. Opt. A Pure Appl. Opt.* 9, 10.1088/1464-4258/9/8/S01 (2007). [23] A. Yao, M. Tassieri, M. Padgett, and J. Cooper, Microrheology with optical tweezers, *Lab Chip* 9, 2568 (2009). [24] S. K. Gupta and M. Guo, Equilibrium and out-of-equilibrium mechanics of living mammalian cytoplasm, *J. Mech. Phys. Solids* 107, 284 (2017). [25] F. Amblard, A. C. Maggs, B. Yurke, A. N. Pargellis, and S. Leibler, Subdiffusion and anomalous local viscoelasticity in actin networks, *Phys. Rev. Lett.* 10.1103/PhysRevLett.77.4470 (1996). [26] A. R. Bausch, F. Ziemann, A. A. Boulbitch, K. Jacobson, and E. Sackmann, Local measurements of viscoelastic parameters of adherent cell surfaces by magnetic bead microrheometry, *Biophys. J.* 10.1016/S0006-3495(98)77646-5 (1998). 28